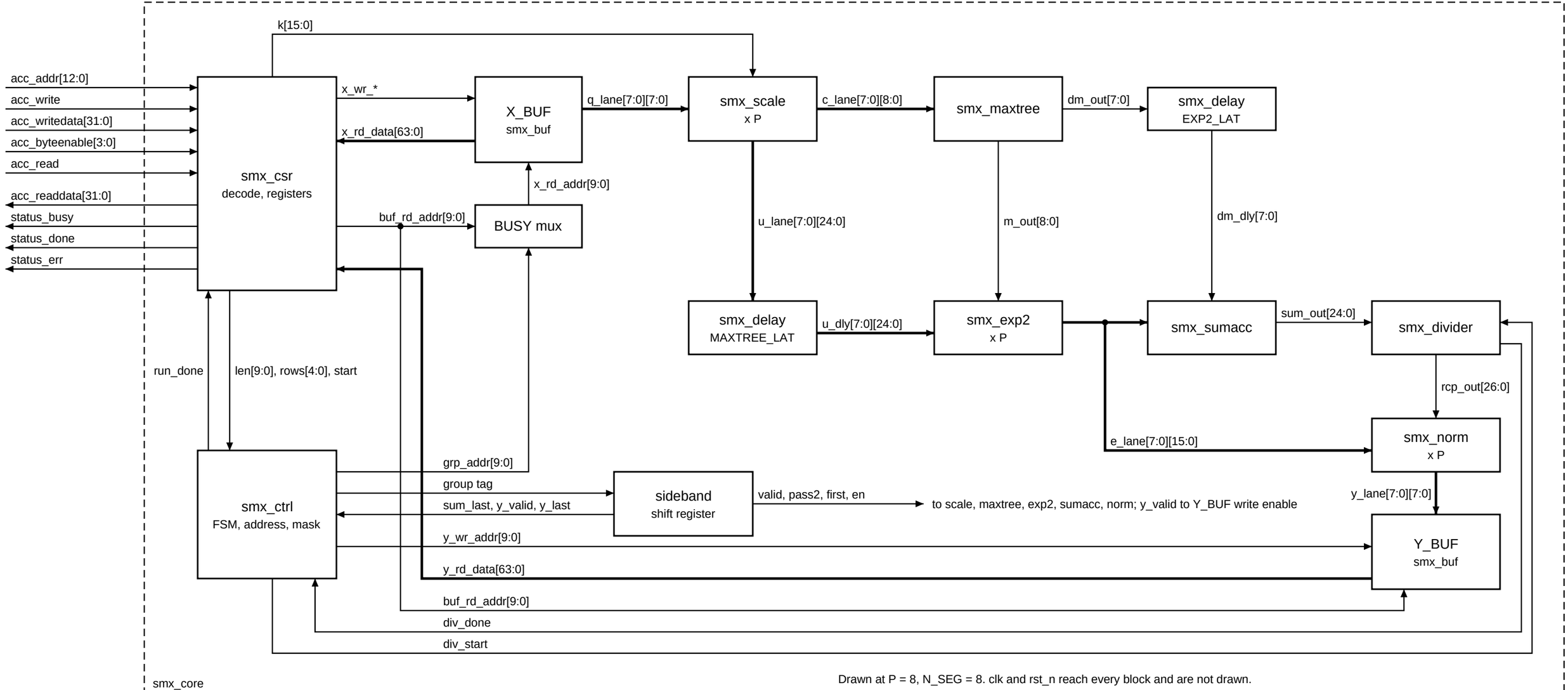
k[15:0]
acc_addr[12:0]
acc_write
acc_writedata[31:0]
acc_byteenable[3:0]
acc_read
acc_readdata[31:0]
status_busy
status_done
status_err
smx_csr
decode, registers
x_wr_*
x_rd_data[63:0]
X_BUF
smx_buf
q_lane[7:0][7:0]
smx_scale
x P
c_lane[7:0][8:0]
smx_maxtree
dm_out[7:0]
smx_delay
EXP2_LAT
x_rd_addr[9:0]
buf_rd_addr[9:0]
BUSY mux
u_lane[7:0][24:0]
m_out[8:0]
dm_dly[7:0]
smx_delay
MAXTREE_LAT
u_dly[7:0][24:0]
smx_exp2
x P
smx_sumacc
sum_out[24:0]
smx_divider
run_done
len[9:0], rows[4:0], start
rcp_out[26:0]
e_lane[7:0][15:0]
smx_norm
x P
grp_addr[9:0]
group tag
sum_last, y_valid, y_last
sideband
shift register
valid, pass2, first, en
to scale, maxtree, exp2, sumacc, norm; y_valid to Y_BUF write enable
y_lane[7:0][7:0]
smx_ctrl
FSM, address, mask
y_wr_addr[9:0]
y_rd_data[63:0]
Y_BUF
smx_buf
buf_rd_addr[9:0]
div_done
div_start
smx_core
Drawn at P = 8, N_SEG = 8. clk and rst_n reach every block and are not drawn.